

temperatures or up to several hundred degrees Celsius. However, operation at extreme temperatures is not true for every semiconductor device or passive component. It depends on a number of materials and design factors.

Limitations of conventional-temperature electronics

Conventional-temperature-range electronics can be used in extreme-temperature environments by means of insulation and heating (for low-temperature environments) or refrigeration (for high-temperature environments). In addition, the electronics may be connected to a thermal sink—a material that can absorb a large amount of

may be less acceptable or less practical than special electronics that can withstand temperature of the environment. Thus operating electronics beyond the normal limits is an option worth considering.

Practical applications of extreme-temperature electronics

Generally speaking, the usual motivation for operating electronics at extreme temperatures is improvement in the overall performance of a system by having part of the system—electronic sub-systems—operate outside the usual temperature range.

The sub-system itself may or may not have improved performance from

The term 'extreme-temperature electronics' refers to electronics operating outside the traditional temperature range of -55°C or -65°C to $+125^{\circ}\text{C}$. It covers very low temperatures, down to essentially absolute zero (0°K or -273°C), as well as high temperatures

heat without a substantial increase in temperature. This is usually done by employing the material's phase change from solid to liquid, which absorbs a large amount of heat (the latent heat of fusion). For low temperature, the opposite effect may be used to provide a thermal source. The same material may serve as a thermal sink for high temperature or a thermal source for low temperature, and may be as basic as ice/water or less familiar such as a bismuth alloy.

However, in many situations, the techniques described above would be undesirable or impractical. There are trade-offs: passive techniques may have a limited lifetime that is insufficient for the application. Active techniques require additional power and sub-systems. For some applications, these may be too disturbing to the environment because of the additional heat that must be dumped. All the techniques add weight and some degree of complexity. These burdens

operating at an extreme temperature. One of the primary reasons is that measurement or control must be done in a cold or hot environment. For example, a major use of HTE is in well-logging. Exploration and monitoring for petroleum and geothermal wells uses probes that are sent deep into bore holes, where the temperature may reach $+200$ to $+300^{\circ}\text{C}$. Since signals from sensors in these probes must be sent to surface instruments via long cables, it is desirable to co-locate some signal-processing electronics in the probes to avoid signal degradation and to multiplex signals. One approach is to allow the electronics to be subjected to the high-temperature environment seen by the probe. This 'hot' electronics in the probe is likely to have degraded performance; however, the overall system of sensors, probe electronics and surface instrumentation will have improved performance compared to a system that has no electronics in the probe.

Practical applications of low-temperature electronics

Electronics is operated at low temperatures due to improved performance of the cooled electronics itself. Improvement upon cooling results from a combination of effects: in general, transistors (field-effect types) exhibit increased gain and speed and lower leakage. Also, parasitic resistances and capacitances in interconnections decrease, heat transfer improves and many devices exhibit lower noise.

There have been two main application areas, viz, speeding up digital electronics and enhancing computer performance. Another successful ongoing area is microwave preamplification. The cooling of amplifiers to reduce noise is well established and this has been employed for many years in the scientific community for receivers used in radio astronomy as well as for deep-space communications with the distant spacecraft. Cooling the transistors greatly reduces their thermal noise, which is dominant at microwave frequencies.

Practical applications of high-temperature electronics

High-temperature electronics (HTE) provides expert coverage of the applications, characteristics, design, selection and operation of electronic devices and circuits at temperatures above the conventional limit of 125°C . Its scope includes active components from standard and advanced semiconductor materials, passive components, as well as technologies for metallisation, interconnections, and the assembly and packaging of electronic components.

At the low-temperature end, practical operation of devices and circuits is reasonably achievable to as low a temperature as desired, provided materials and designs appropriate to the temperature are used. The various characteristics of a device may improve or degrade. In particular, below about -230°C (40°K) Si devices often exhibit